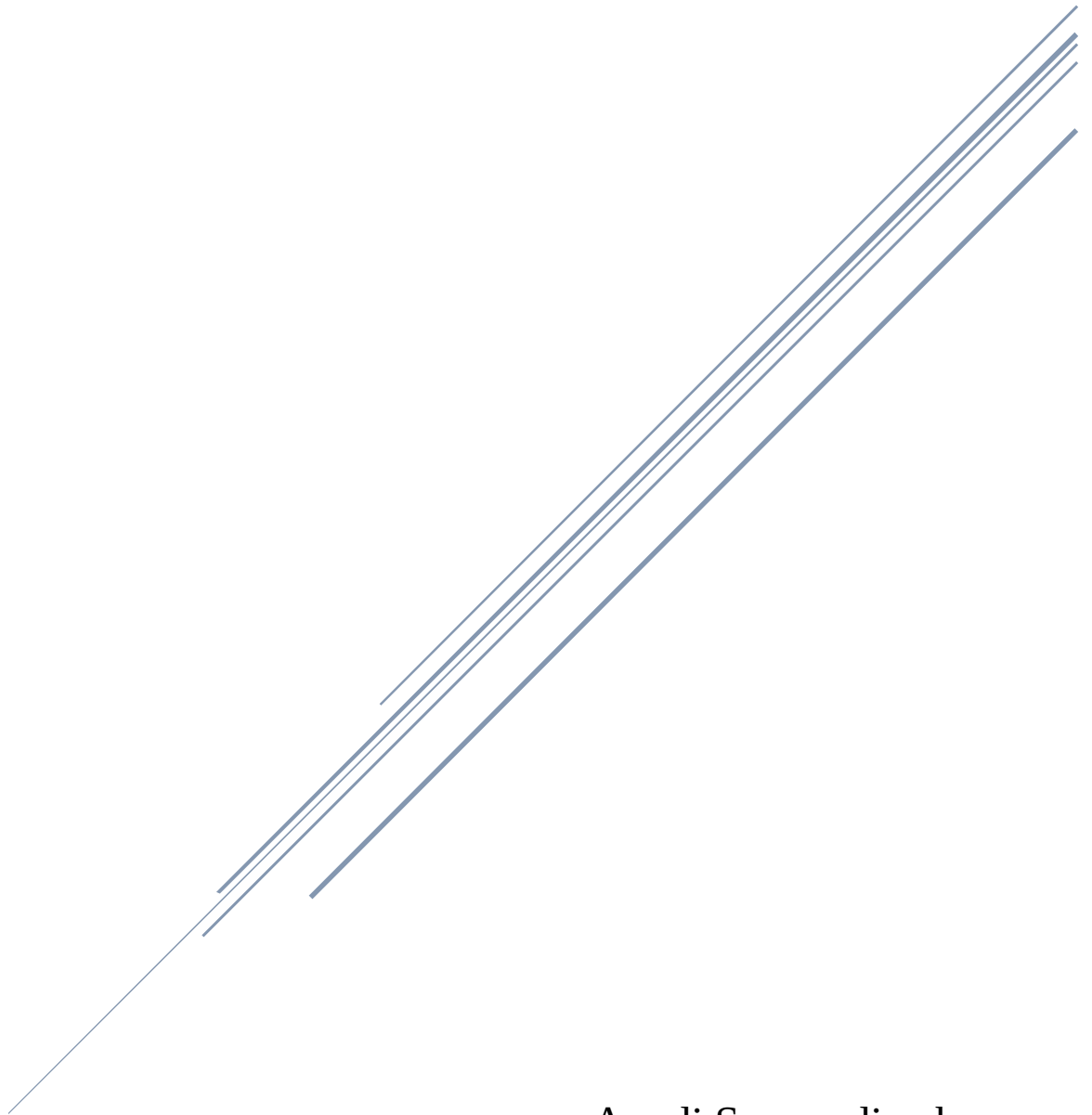


Phishing Awareness Analysis

DecodeLabs Cyber Security Internship – Project 3



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Abstract

Phishing is one of the most common cyberattacks used by criminals to steal confidential information from individuals and organizations. Attackers often disguise themselves as trusted entities and send fraudulent emails or messages to deceive users. The objective of this project is to identify phishing attempts by analyzing suspicious emails and comparing them with legitimate communications.

This project focuses on recognizing phishing indicators, identifying suspicious links, understanding social engineering techniques, and increasing cybersecurity awareness.

Introduction

Cybercriminals use various methods to gain unauthorized access to sensitive information. Among these methods, phishing remains one of the most dangerous and widely used attacks. These attacks rely on human error rather than technical vulnerabilities.

Phishing attacks usually include:

- Fake sender addresses
- Suspicious hyperlinks
- Requests for sensitive information
- Threatening or urgent language
- Spelling and grammatical errors

Understanding these indicators is essential for maintaining information security.

Project Objectives

The objectives of this project are as follows:

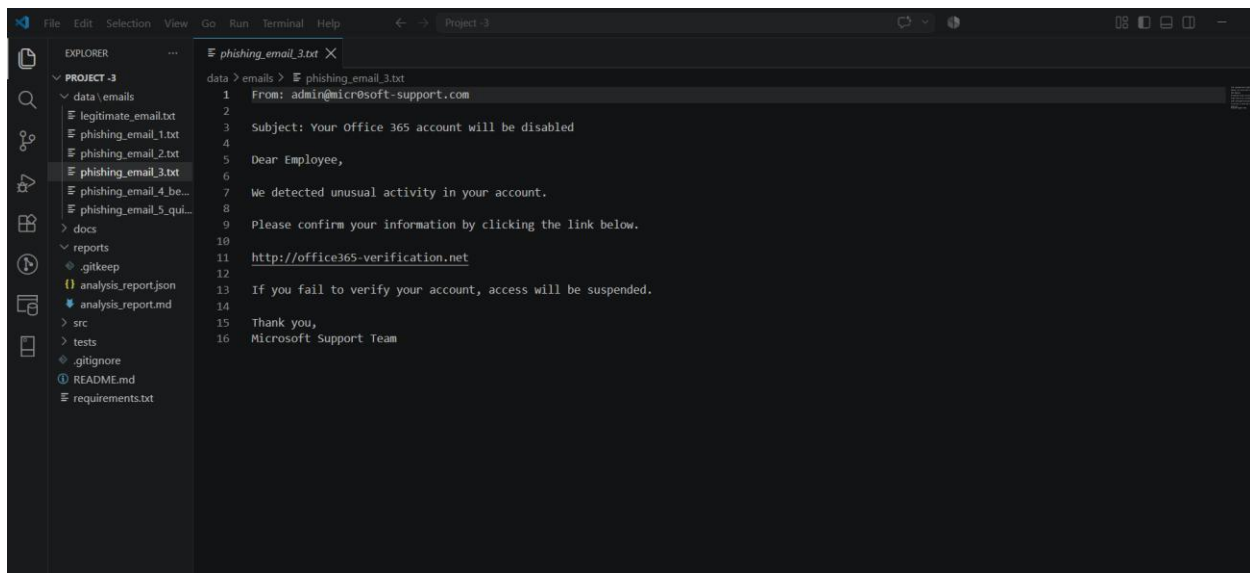
- Analyze suspicious emails.
- Identify phishing indicators.
- Detect malicious links.
- Distinguish legitimate emails from fraudulent messages.
- Improve cybersecurity awareness.

This screenshot shows the Visual Studio Code interface with a project named 'PROJECT -3'. The Explorer sidebar on the left shows a file tree with folders 'data', 'docs', 'reports', 'src', and 'tests'. The 'data' folder is expanded, showing a subfolder 'emails' containing several text files, including 'phishing_email_1.txt' which is currently selected. The main editor window displays the content of 'phishing_email_1.txt'. The email text is as follows:

```
data > emails > phishing_email_1.txt
1 From: support@paypal.com
2
3 Subject: Urgent Action Required
4
5 Dear Customer,
6
7 Your account has been suspended. Click the link below immediately to verify your account.
8
9 http://paypal-security-login.com
10
11 Failure to act within 24 hours will result in account termination.
12
13 Thank you,
14 Support Team
```

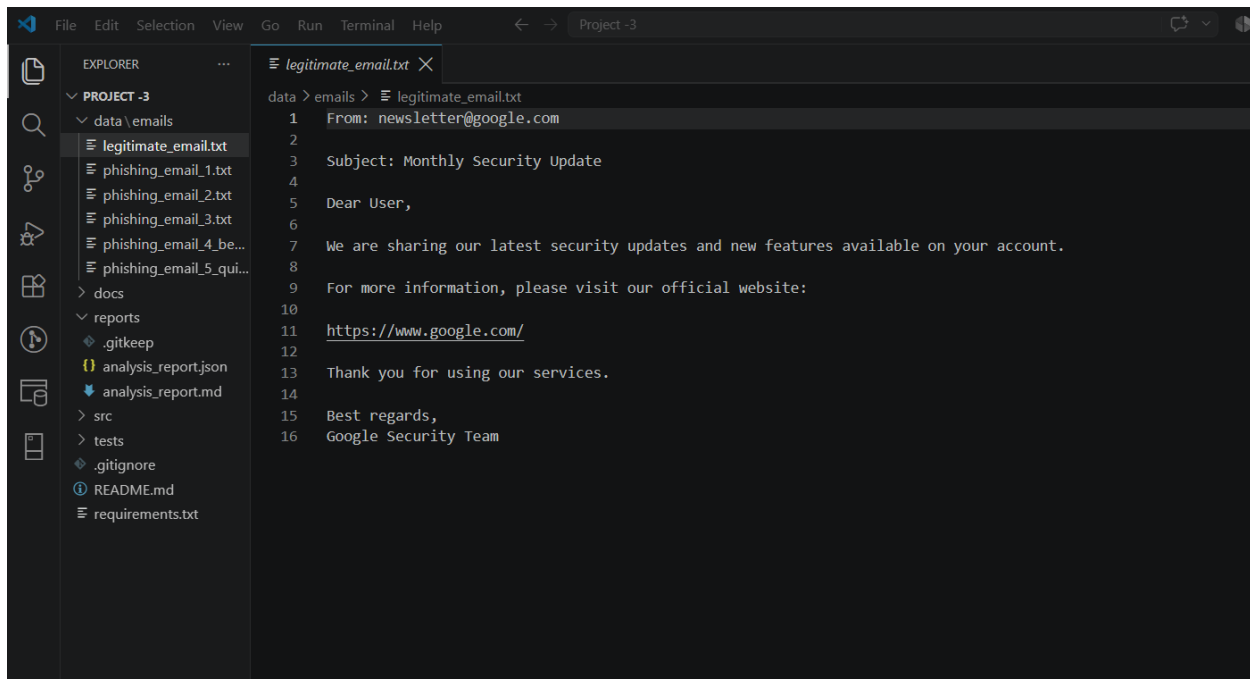
This screenshot shows the Visual Studio Code interface with the same 'PROJECT -3' project. The Explorer sidebar shows the 'emails' folder expanded, with 'phishing_email_2.txt' selected. The main editor window displays the content of 'phishing_email_2.txt'. The email text is as follows:

```
data > emails > phishing_email_2.txt
1 From: rewards@amazon-gifts.com
2
3 Subject: Congratulations! You have won a prize.
4
5 Dear User,
6
7 You have been selected to receive a free smartphone.
8
9 To claim your reward, click the link below and provide your personal details.
10
11 http://amazon-reward-center.com
12
13 This offer expires in 12 hours.
14
15 Congratulations,
16 Reward Team
```



A screenshot of the Visual Studio Code editor interface. The Explorer sidebar on the left shows a project structure with folders like 'data', 'reports', and 'src'. The file 'phishing_email_3.txt' is selected. The main editor window displays the content of this file, which is a phishing email from 'admin@microsoft-support.com' with a subject line 'Subject: Your Office 365 account will be disabled'. The email body contains a warning about unusual activity and a link to 'http://office365-verification.net'.

```
1 From: admin@microsoft-support.com
2
3 Subject: Your Office 365 account will be disabled
4
5 Dear Employee,
6
7 We detected unusual activity in your account.
8
9 Please confirm your information by clicking the link below.
10
11 http://office365-verification.net
12
13 If you fail to verify your account, access will be suspended.
14
15 Thank you,
16 Microsoft Support Team
```

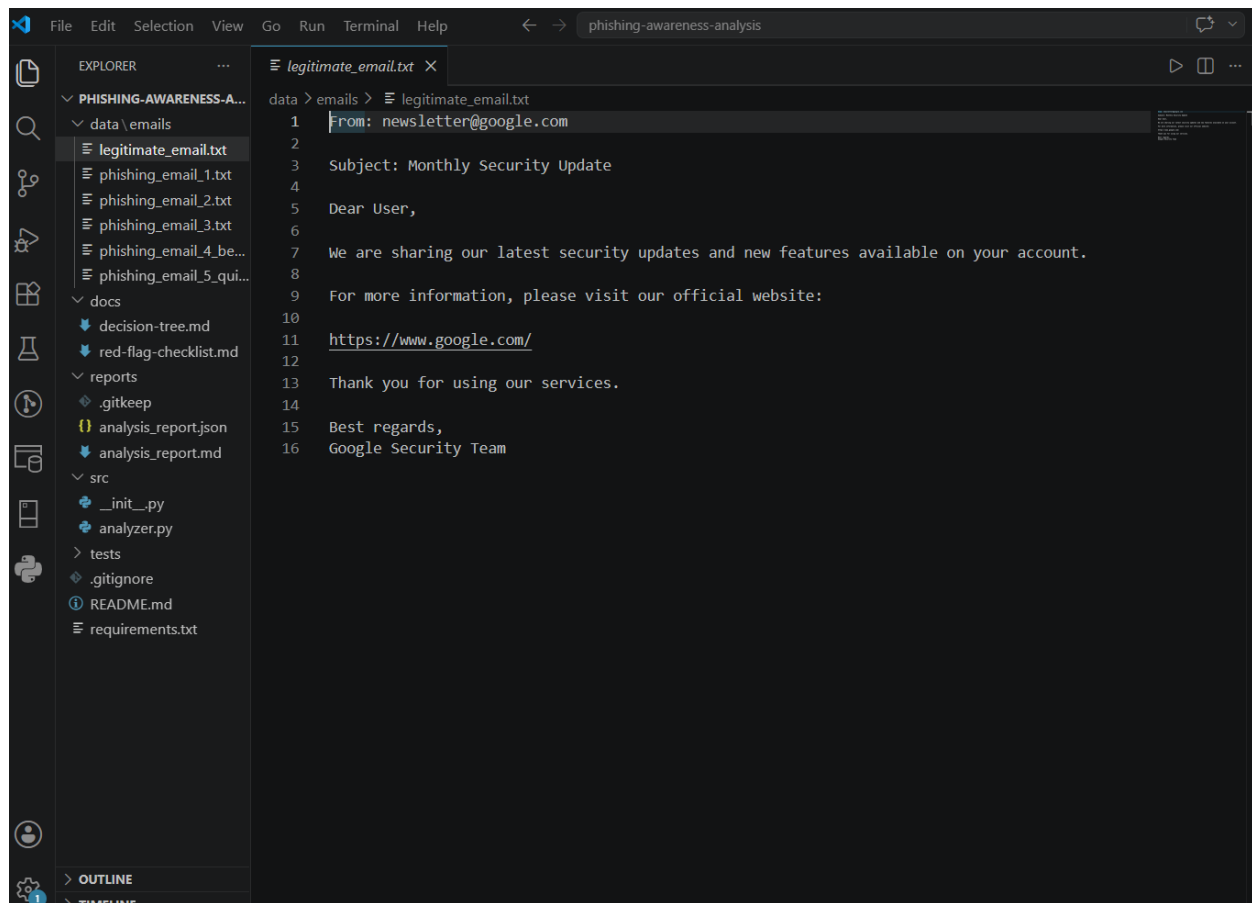


A screenshot of the Visual Studio Code editor interface. The Explorer sidebar on the left shows the same project structure. The file 'legitimate_email.txt' is selected. The main editor window displays the content of this file, which is a legitimate email from 'newsletter@google.com' with a subject line 'Subject: Monthly Security Update'. The email body contains information about security updates and a link to 'https://www.google.com/'.

```
1 From: newsletter@google.com
2
3 Subject: Monthly Security Update
4
5 Dear User,
6
7 We are sharing our latest security updates and new features available on your account.
8
9 For more information, please visit our official website:
10
11 https://www.google.com/
12
13 Thank you for using our services.
14
15 Best regards,
16 Google Security Team
```

```
1 [{
2   {
3     "filename": "legitimate_email.txt",
4     "sender_display_name": "",
5     "sender_email": "newsletter@google.com",
6     "subject": "Monthly Security Update",
7     "urls": [
8       "https://www.google.com/"
9     ],
10    "red_flags": [],
11    "score": 0,
12    "verdict": "SAFE",
13    "action": "Close"
14  },
15  {
16    "filename": "phishing_email_1.txt",
17    "sender_display_name": "",
18    "sender_email": "support@paypal.com",
19    "subject": "Urgent Action Required",
20    "urls": [
21      "http://paypal-security-login.com"
22    ],
23    "red_flags": [
24      {
25        "code": "BRAND_DOMAIN_MISMATCH",
26        "severity": "high",
27        "message": "Message references brand 'paypal' but sender domain 'paypal.com' does not match any official domain (paypal.co",
28      },
29      {
30        "code": "LOOKALIKE_DOMAIN",
31        "severity": "high",
32        "message": "Sender domain 'paypal.com' looks like a disguised version of 'paypal.com' (character substitution / typosquat",
33      },
34      {
35        "code": "NON_HTTPS_URL",
```

```
1 [{
2   {
3     "filename": "legitimate_email.txt",
4     "sender_display_name": "",
5     "sender_email": "newsletter@google.com",
6     "subject": "Monthly Security Update",
7     "urls": [
8       "https://www.google.com/"
9     ],
10    "red_flags": [],
11    "score": 0,
12    "verdict": "SAFE",
13    "action": "Close"
14  },
15  {
16    "filename": "phishing_email_1.txt",
17    "sender_display_name": "",
18    "sender_email": "support@paypal.com",
19    "subject": "Urgent Action Required",
20    "urls": [
21      "http://paypal-security-login.com"
22    ],
23    "red_flags": [
24      {
25        "code": "BRAND_DOMAIN_MISMATCH",
26        "severity": "high",
27        "message": "Message references brand 'paypal' but sender domain 'paypal.com' does not match any official domain (paypal.co",
28      },
29      {
30        "code": "LOOKALIKE_DOMAIN",
31        "severity": "high",
32        "message": "Sender domain 'paypal.com' looks like a disguised version of 'paypal.com' (character substitution / typosquat",
33      },
34      {
35        "code": "NON_HTTPS_URL",
```



Tools Used

- Microsoft Word
- Notepad
- Virus Total
- URLScan.io
- Snipping Tool
- GitHub

Skills Acquired

- Threat analysis
- Risk assessment
- Cybersecurity awareness
- Social engineering analysis
- Email security analysis

Results

The project successfully identified multiple phishing attempts and highlighted the techniques used by attackers to deceive users. By comparing legitimate and fraudulent emails, it became easier to understand how cybercriminals manipulate human behavior.

Conclusion

This project provided practical experience in recognizing phishing attacks and understanding their potential impact. The knowledge gained from this project can help users protect themselves from malicious activities and strengthen their overall cybersecurity awareness.

Continuous learning and careful analysis are essential for identifying and preventing future attacks.